

Recursive-CSD N x N unitary synthesis -- CONSTRUCTED optimal T-count vs N
Xinyu Tan, 'Unitary synthesis with fewer T gates', arXiv:2509.25702.

Algorithm: recursive cosine-sine decomposition factors $U(2^n)$ into $2^n - 1$ multi-controlled single-qubit unitaries (Thm 3.2/3.5). Consecutive runs of $2^k - 1$ of them that share the same k-qubit target are GROUPED into a single $(n-k)$ -controlled k-qubit unitary; there are 2^{n-k} such blocks. Each is synthesized by a generalized Gosset-Kothari-Wu diagonal synthesis at T-count $2^{(n+k)/2} \sqrt{L} + 4^k L$ (Thm 4.3). Summing over 2^{n-k} blocks and minimizing over k (asymptotic optimum $k \sim n/3$):

$$T = O(2^{4n/3} L^{2/3}) \sim N^{4/3} \quad \text{up to logs} \quad (\text{Thm 1.1}).$$

These T-counts are READ OFF THE ACTUAL CIRCUIT, not the formula: each point is the Qualtran QECGatesCost of PaperRecursiveCSDUnitarySynthesis = 2^{n-k} constructed MultiControlledKQubitUnitaryQROAM blocks (a QROAM select-swap lookup over the n-k controls loading 4^k phase words + 4^k phase-gradient additions on the k targets). k is chosen to minimize the real gate count at each n; phase word length $b = 32$.

Block-diagonal $U = \sum_a |a\rangle\langle a| (x) U_a$ (here $N_k = 216 = 6^3$): the block label a is appended to every QROAM lookup address (table $(N_k, 2^{n-k})$), so only the lookup term grows -- by $\sim \sqrt{N_k}$ under the select-swap optimum -- while the 4^k reconstruction is unchanged. The optimal k therefore shifts UP for the block case, and the N-scaling exponent is unchanged ($\sqrt{N_k}$ is a constant prefactor): both cases approach 4/3.

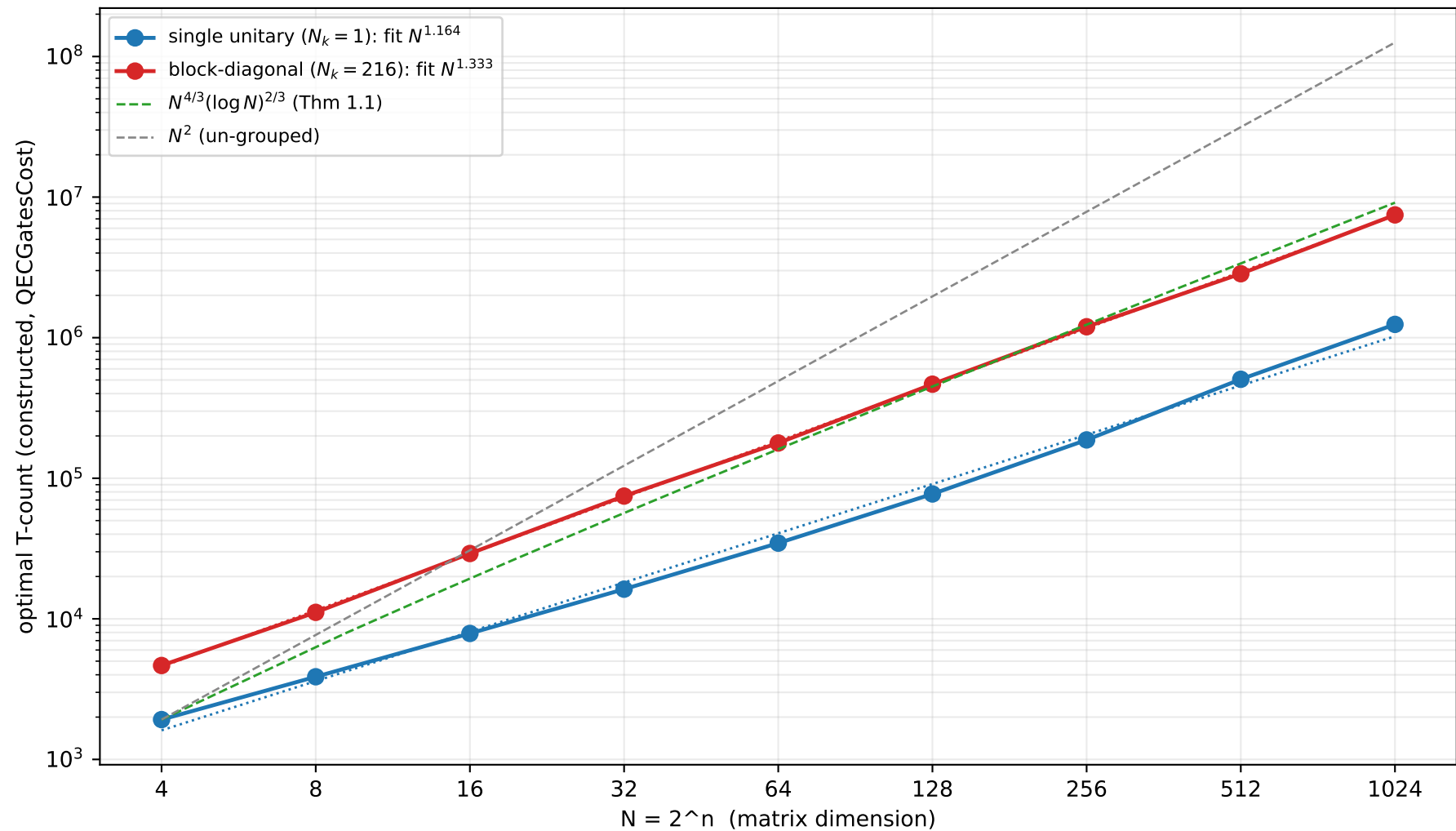
Power-law fits over $N = 4..1024$:

single ($N_k=1$): $T \sim 3.207e+02 * N^{1.164}$

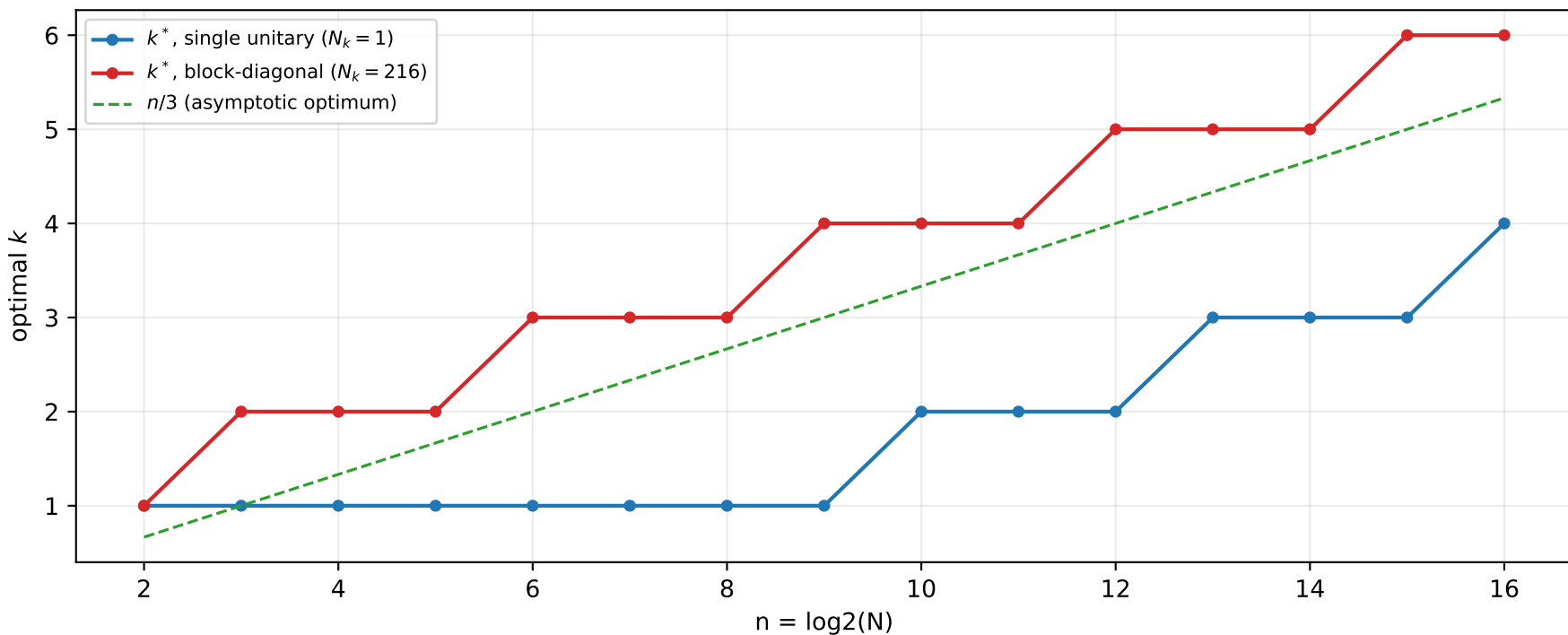
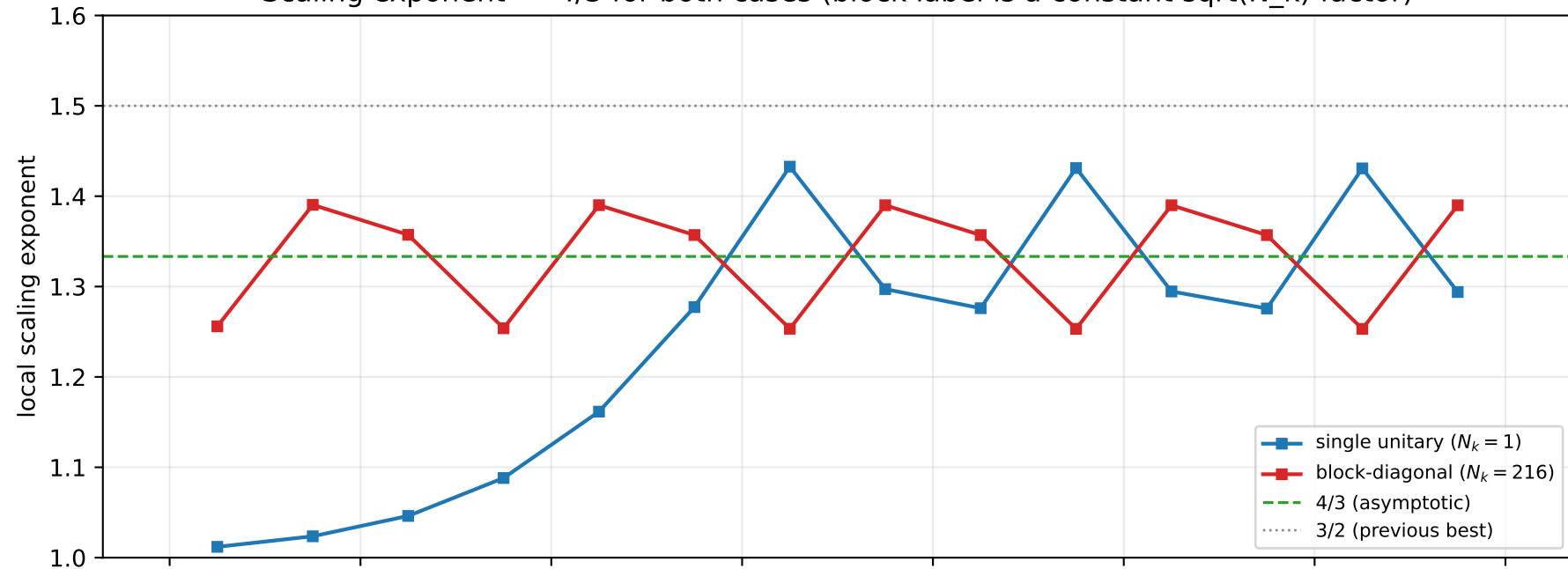
block ($N_k=216$): $T \sim 7.198e+02 * N^{1.333}$

The fitted exponent is below 4/3 in this finite range (small n keeps k small); the constructed local exponent climbs toward 4/3 as n grows (asymptotic page) with $k^* \rightarrow n/3$. Both are far below the N^2 cost of the explicit un-grouped circuit (RecursiveCSDSynthesisQROAM).

Optimal T-count vs N: single vs block-diagonal ($N_k = 216$), $b=32$



Scaling exponent $\rightarrow 4/3$ for both cases (block label is a constant $\sqrt{N_k}$ factor)



Constructed optimal T-count: single vs $N_k=216$ block-diagonal ($b=32$)

n	N	k^* ($Nk=1$)	T ($Nk=1$)	k^* ($Nk=216$)	T ($Nk=216$)	ratio
2	4	1	1,920	1	4,656	2.4
3	8	1	3,872	2	11,120	2.9
4	16	1	7,872	2	29,152	3.7
5	32	1	16,256	2	74,688	4.6
6	64	1	34,560	3	178,112	5.2
7	128	1	77,312	3	466,816	6.0
8	256	1	187,392	3	1,195,776	6.4
9	512	1	505,856	4	2,850,560	5.6
10	1024	2	1,243,136	4	7,470,592	6.0